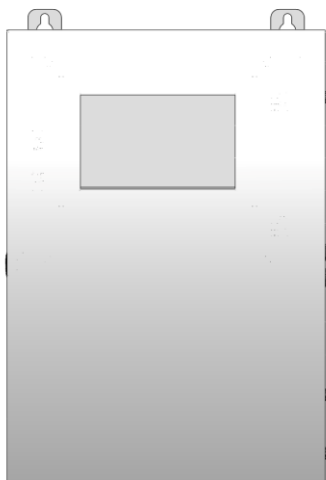
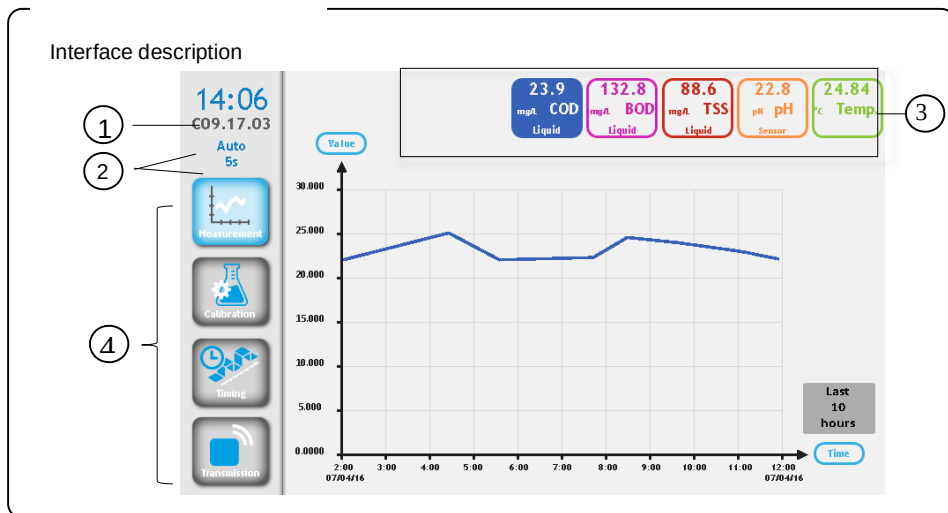




Standard Operation Procedures

Display

Hemera on-line analyzers are all supplied with an easy-to-use colour touch screen display interface. The interface allows the user to easily navigate through a multitude of screens that are used to set and check all the operating conditions of the instrument. Once you turn on the analyser, you can get the below interface:



Hemera analyzers can be single parameter or multi-parameter analyzers.

1- Time / Date:

When you click on this field, you can adjust the date and the clock of analyser.

2- Measurement info:

In this area, you have information about software version and analyser status

3- Parameters:

You can find the list of parameter. As shown on the example above the analyzer measures 5 parameters: COD, BOD, TSS, pH and Temperature.

4- Main Menu:

In this area, you have 4 buttons corresponding to main menu: Measurement, Calibration and Transmission.

When the colour of a button be BLUE, it means the function has been selected, such as:



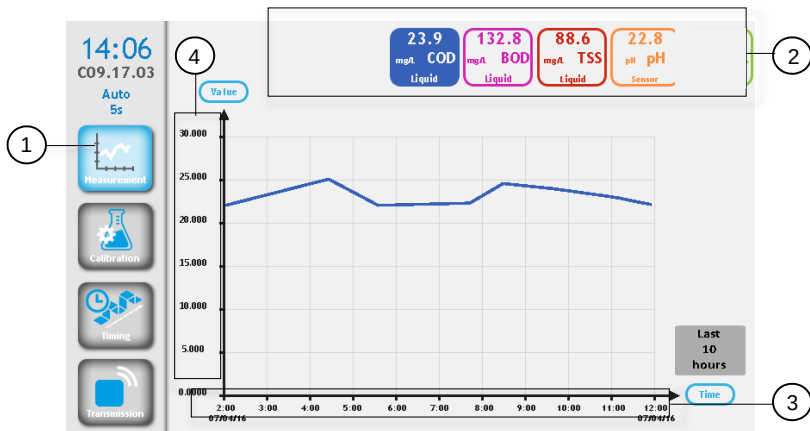
When the colour of a button be GREY, it means the function has been selected, such as button.

The following chapter 8,9,10 and 11, have explained more details about each function of

Measurement interface

This screen displays historic of measurement.

Measurement menu



1 - Measurement menu

This screen displays last concentration reading and graph of the measured data over time for each parameter selected on the top.

2 - Parameters

List of parameters measured by the analyzer. You can select parameters that you want to display historic measurements graph.

When the parameter button has full colour, it means be selected:



Otherwise, it means not be selected:



3 - Time

It shows the time of measurement.

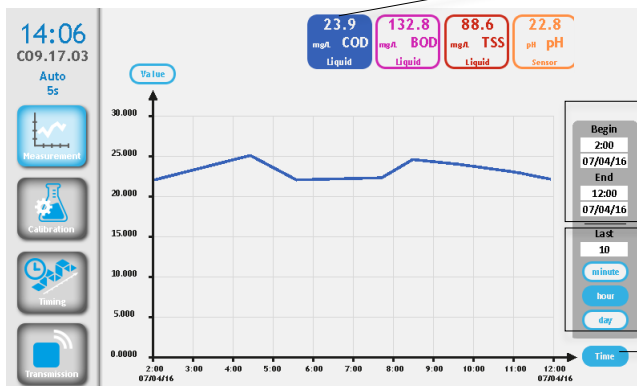
4 - Value

You can zoom the result by changing the scale of Y-axis value.

Adjustment of time scale

On the right side, you can adjust the start and the end of the historic that you want to display on graph. Or you can select the last 10 minutes/hours/days as well. Follow the point 1, 2, 3 in the below figure:

Historic scale



1. Click here to choose the parameters you want to see for its history.

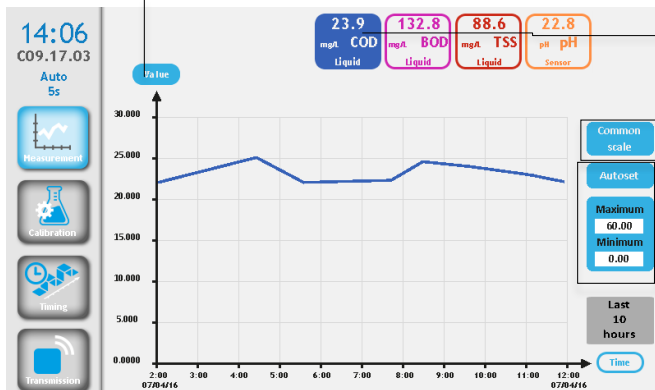
3. Entry here for the BEGIN and END of period you want to select.

3. Or choose the last how many minutes/hour/day.

2. Click here to adjust the time scale

Adjustment of measuring range

On the right side, you can adjust if you want a common scale for all parameters, or if you want an auto set, or if you want to set maximum and minimum values. Follow the point 1, 2, 3, 4 in the below figure:



2. Click here to adjust the measuring range

1. Click here to choose the parameters you want to see for its history.

3. Click common scale if you want all parameters have the same scale. If not deselect it, and each parameter has its own scale.

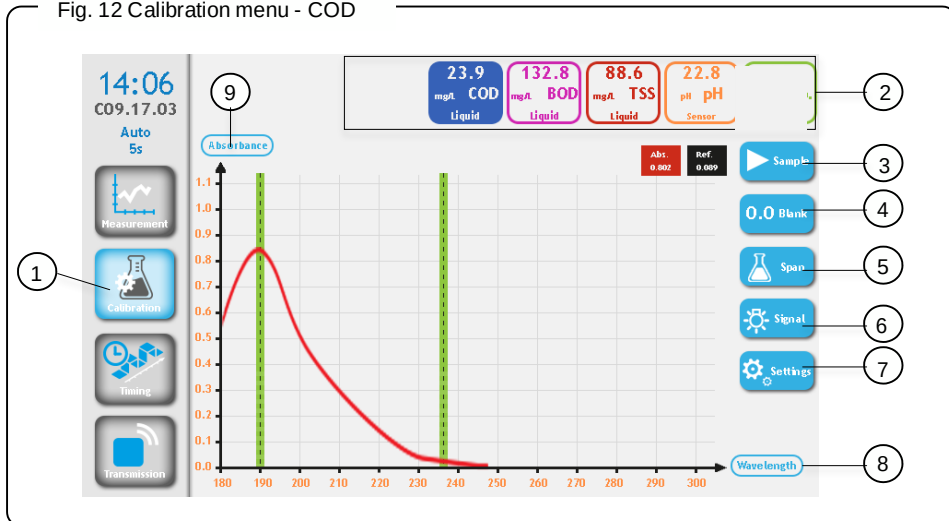
4. Entering the maximum and minimum of scale you want for parameter or parameter.

Range scale

Manual checking

This screen displays parameter selected of the measured sample. It allows also perform manual measurement and calibration if periodic measurement and zeroing is not activated. To make a manual measurement or to set the calibration, you have to select the parameters that you want to adjust. This field displays the name of the parameter and measured value.

Fig. 12 Calibration menu - COD



1 - Calibration menu

This screen displays absorption spectrum of the measured sample on selected wavelength. It allows also perform manual measurement and calibration.

2 - Parameters

To make a manual measurement or to set the calibration, you have to select the parameters that you want to adjust. This field displays the name of the parameter and measured value.

3 - Sample

The manual measurement works once periodic measurement is not activated.

4 - Blank

The manual zeroing works once periodic measurement and zeroing is not activated.

5 - Span

Activate manual calibration.

6 - Signal menu

Go to spectrograph calibration menu.

7 - Settings menu

Go to parameters configuration menu.

8 - Wavelength scale

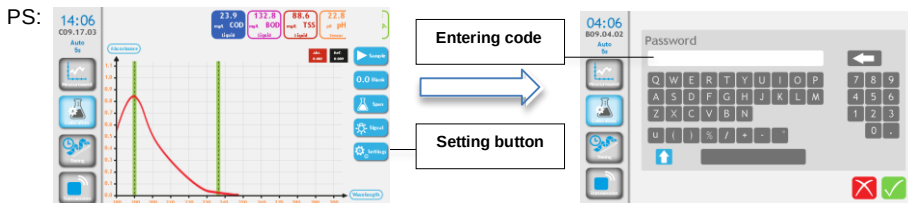
Adjust of wavelength range.

9 - Absorbance scale

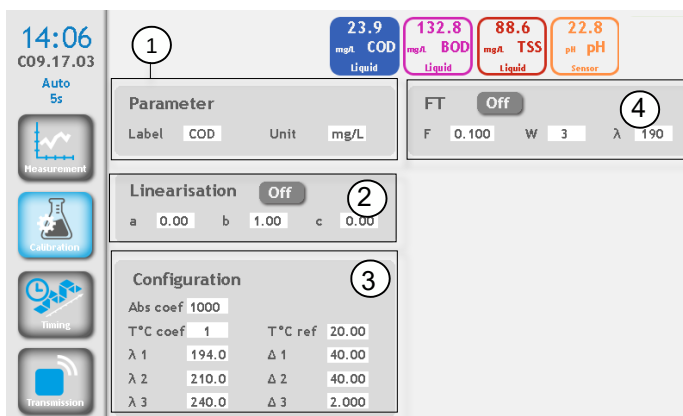
Adjust of Absorbance scale.

Settings menu

This screen allows modifying only some configuration of the parameters but most of them are already set in factory. Depending of the application, different mathematical treatment is activated as you can see on test report provided with the analyzer.



Setting menu



1 - Parameter

This area of the screen allows to change label, unit and digits.

2 - Linearization

This area of the screen allows activating or not a linearization curve. You have to enter the coefficient of first, second and third degrees.

3 - Configuration

This area of the screen allows adjusting value set in factory

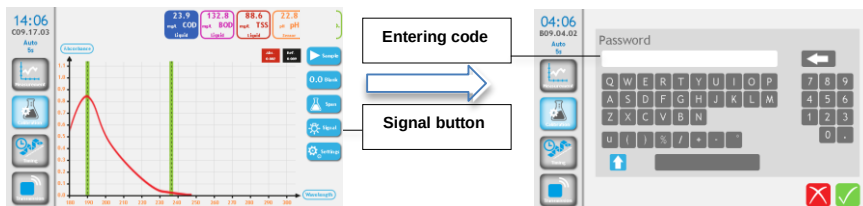
4 - FT algorithm

This area of the screen allows activating or not the FFT treatment.

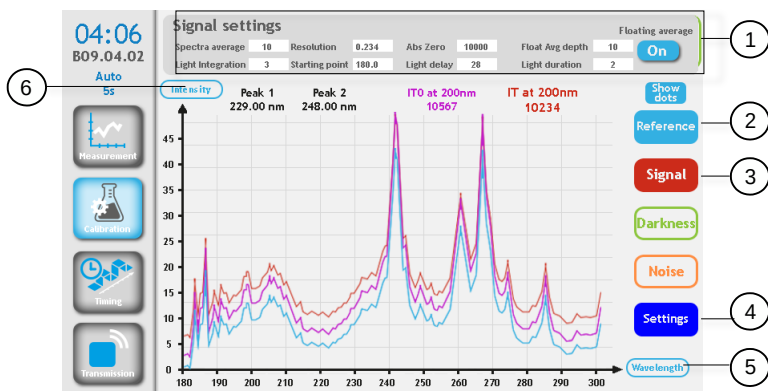
Signal menu

This screen helps to adjust and set the spectrograph, in this analyser, there's just COD and HC (Oil in water) have SIGNAL, they are measured based on spectroscopy.

PS:



Signal menu



1 - Signal settings

You can calibrate ("resolution" & "starting point") and adjust the quantity of light you need to get appropriate zero on your spectrograph. This can be done with "light integration". In order to get stable measurement, you can increase the number in "spectra average".

2 - Reference

You can keep in memory a reference signal, which is the maximum signal you have when the flow cell is clean. By comparing IT₀ to this reference signal, you can observe the lost of transmission due to deposit on the windows.

3 - Signal

You can check the signal transmission manually.

4 – Settings

You can display or not "signal settings" menu

5 – Wavelength scale

Adjust of Wavelength scale.

6 – Intensity scale

Adjust of Intensity scale.

Chronogram

This screen helps to set measurement process and periodicity of zeroing and measurement.

Timing menu

1 - Timing

This screen allows setting measurement and zeroing in automatic mode, and it checks the electric valves and pumps activation.

2 - Automatic mode

The analyzer can be operated in automatic mode. Pressing ON activates automatic measurements. The frequency of measurement can be set in seconds, minutes or hours by entering the value of the frequency.

You can activate the storage of measure on SD card or not.

An autocalibration can be activated. The frequency of auto calibration can be set by number of zeroing. You have to indicate the span concentration, the error in percentage that you accept and the flow time of span in the flow cell.

An automatic cleaning cycle can be activated by pressing ON. The frequency of cleaning cycle can be set by number of measurement.

Pressing ON can also activate an automatic zeroing.

3 - Measuring process

This field allows setting the sequences of measurement cycle, which are split in 5 steps (Zeroing (for liquid or gas sample), Sampling, Reaction, Measurement (for liquid or gas sample) and Purge). Time is in second.

4 - Pumps and valves control

This field allows activation of electric valves and pumps and status during measurement cycle.

5 - Minimum measurement Frequency

This field displays the minimum frequency between 2 measurements, the acquisition time for liquid measurement and the acquisition time for gas measurement.

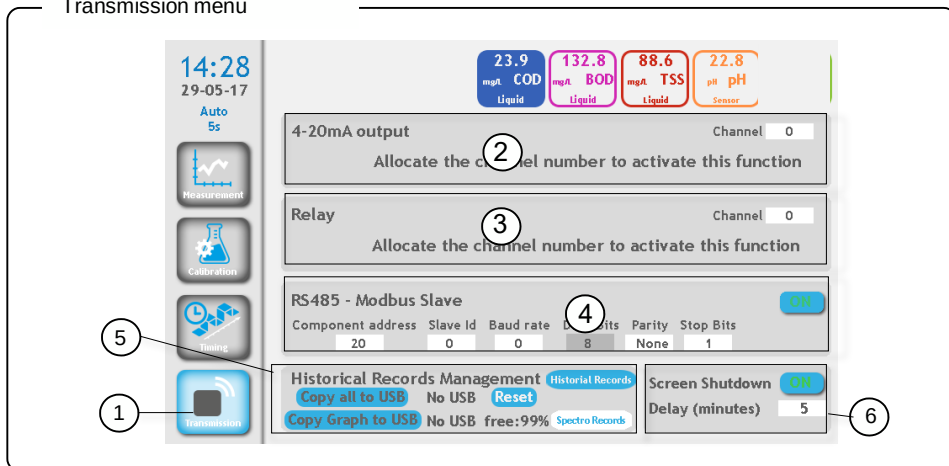
6 - PID control

This field allows setting the temperature of oven and the PID for the stabilisation.

Transmission or communication

Communication can be done via analogue and digital outputs as describe below.

Transmission menu



1 - Transmission menu

This screen is used to set the communication mode.

2 - 4-20mA outputs

The analyzer is supplied with up to 4x 4-20 mA outputs. The range can be set for each parameter. The current milliamp signal corresponds to the last measurement. This value will be updated after each measurement.

3 - Relays

The analyzer is supplied with up to 4x Relays. Threshold high or low alarm value can be set for each parameter.

4 - RS485

The analyzer can communicate by Modbus via RS485 connection. This field gives information about addresses.

5 - USB

The analyzer can transfer diagnostic data and measurement data stored in .txt files in USB key. On this field you can find information on software version and free memory space.

6 - Automatic Screen Shutdown

The analyzer has a screen saver option. You can be activated this option on this field and set the delay in minute.